

Superposed: A Quantum Factory Puzzle Game

Abstract

Superposed is a concept for an automation puzzle game in which players build and manage a factory floor where the “products” are quantum states. Rather than manipulating geometric shapes, players route qubits through a network of quantum gates, learning the fundamental behaviours of quantum information - superposition, measurement, and interference - through hands-on experimentation.

The game requires no prior knowledge of physics or mathematics. All concepts are introduced gradually through visual feedback and puzzle constraints. The goal is to build lasting intuition about quantum computing before any formal notation is encountered.

1 Introduction

Superposed is a puzzle automation game inspired by factory-building games such as *Shapez*, translating the same intuitive mechanics into the domain of quantum computing.

Players construct factories that process qubits instead of physical objects. The design philosophy avoids equations and formalism entirely: quantum concepts emerge naturally through interaction and experimentation rather than instruction.

The target audience consists of complete beginners and curious non-specialists. The ultimate goal is for players to develop a durable mental model of how quantum computation behaves.

2 Game State and Entities

The game world consists of a factory floor populated by interacting entities:

2.1 Qubits

Qubits are the primary “products” of the factory. Their state is represented visually:

- Red: $|0\rangle$
- Blue: $|1\rangle$
- Purple: superposition

Players never see formal representations such as the Bloch sphere. Instead, intuition is built through visual metaphors.

2.2 Factory Components

- **Conveyor belts**: transport qubits between components.
- **Quantum gates**: transform qubits. Examples include:
 - Hadamard: creates superposition
 - X gate: flips states

- CNOT: entangles two qubits
- **Measurement nodes:** collapse superposition probabilistically.
- **Splitters:** duplicate classical routing paths (not quantum states).
- **Output sinks:** define puzzle goals.

2.3 Extended Entities

- **Black box machines:** opaque systems whose internal behaviour is hidden.
- **Oracle locks:** puzzle gates that accept only a specific (unknown) quantum state.

3 Player Actions

Players interact with the factory through a set of simple but expressive actions:

- Place and remove gates on the grid
- Draw, redirect, split, and merge conveyor belts
- Rotate gate orientation (important for multi-qubit gates)
- Pause, run, and step through simulation
- Inspect qubits without collapsing their state (game abstraction)

Additional mechanics extend interaction:

- Send qubits into black box machines to observe outputs
- Probe oracle locks using custom test circuits

4 Core Mechanics

4.1 Black Box Testing

Black box machines hide an unknown transformation. Players must infer its behaviour by sending test inputs and observing outputs.

- Limited number of probes per level
- Encourages strategic experimentation
- Successful identification unlocks the gate

This mechanic mirrors real-world quantum algorithm design, where unknown functions are accessed only through queries.

4.2 Oracle Locked Levels

Oracle puzzles require players to determine which quantum state unlocks a level.

1. Observe visual hints
2. Build a probing circuit
3. Analyze outputs
4. Produce the correct state

The oracle acts as a function that can be queried but not inspected, reflecting its role in quantum algorithms.

5 Rules and Constraints

The game enforces rules derived from quantum mechanics:

- **No cloning:** qubits cannot be duplicated
- **Measurement is destructive:** superposition collapses irreversibly
- **Entanglement constraint:** linked qubits must remain synchronized
- **Limited probing:** black box interactions are restricted
- **Oracle opacity:** oracle internals cannot be inspected
- **Gate budget:** encourages efficient design

These constraints are presented purely as gameplay rules rather than formal theory.

6 Game Space

The factory exists on an infinite 2D grid that expands over time.

- Early puzzles restrict space to small regions
- Later puzzles allow large-scale designs
- Each level is a bounded sub-region with fixed inputs and outputs

The possibility space evolves as new gates are unlocked. Black boxes and oracles act as constraints that shape player solutions.

A sandbox mode allows unrestricted experimentation after completing the campaign.

7 Scenario Worlds

The game is structured into four worlds, each introducing one core concept:

7.1 Tutorial World

Single-qubit gates only. Players learn basic transformations and measurement.

7.2 Entanglement World

Introduces CNOT and multi-qubit interactions. Players produce entangled states.

7.3 Interference World

Focuses on phase and interference. Some behaviours are only observable with superposition inputs.

7.4 Algorithm World

Players construct circuits that resemble real quantum algorithms such as Deutsch’s algorithm and quantum teleportation.

Black box and oracle mechanics scale in complexity across all worlds.

8 Development Roadmap

8.1 Phase 1: Core Prototype

Implement basic mechanics: qubits, belts, X and Hadamard gates, and measurement.

8.2 Phase 2: Tutorial World

Establish visual language and introduce black box mechanics.

8.3 Phase 3: Advanced Mechanics

Add entanglement, interference, and oracle puzzles.

8.4 Phase 4: Full Release

Complete algorithm world, sandbox mode, and polish.

9 Pedagogical Insight

A key design insight is that players learn quantum behaviour implicitly through interaction.

In particular:

- Black box testing teaches reasoning about unknown systems
- Oracle puzzles introduce the concept of query-based computation
- Interference puzzles demonstrate that superposition enables fundamentally different outcomes

A notable moment occurs when players discover that certain systems cannot be distinguished using classical inputs alone, revealing the power of quantum computation through gameplay.

10 Conclusion

Superposed aims to bridge the gap between abstract quantum theory and intuitive understanding. By embedding authentic quantum principles into an accessible factory-building framework, the game enables players to develop a meaningful mental model of quantum computation without formal mathematics.